

Supplementary Material: Exact Wave-Function Solution and Spurious-Mode Discrimination for Completely Free Annular Sector Plates

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This file is the complete Supplementary Material to the companion manuscript: §§S.1–S.4, Tables S.1–S.11, and the three references cited below. Equation tags (B.1), (B.2), (B.3) and (B.3') retain the source-paper numbering. Main-text §3.3 cites (B.1), (B.2) and (B.3'); (B.3) appears only here.

S.1 Frobenius series construction

Following Ref. [1] §3 (out-of-plane) and Ref. [2] §3 (in-plane), the radial ordinary differential equation is expanded in Frobenius series about the sector's center radius r_0 (equivalently $r^* = 0$) rather than about the singular point $r = 0$, so that all four independent solutions remain regular on $r^* \in [-\pi/2, \pi/2]$:

$$G_q(r^*, \xi) = \sum_{m=0}^M a_m^{*(q)}(\xi) r^{*\lambda(q)+m}, \quad q = 1, \dots, 4 \quad (\text{Ref. [1] Eq. (32)}).$$

The indicial equation,

$$\bar{r}_0^4 a_0^* \lambda(\lambda - 1)(\lambda - 2)(\lambda - 3) = 0 \quad (\text{Ref. [1] Eq. (31)}),$$

fixes $\lambda(q) = q - 1$ for $q = 1, \dots, 4$, giving the four out-of-plane branches their characteristic near-center behavior $(r^{*0}, r^{*1}, r^{*2}, r^{*3})$. The paper states the resulting recursion for the coefficients $a_m^{*(q)}$ explicitly only through this indicial step, noting it is “much too cumbersome to include” in closed form (Ref. [1] §3); the present implementation carries the recursion numerically to convergence at working precision (`mpmath`, 40 digits) for each candidate frequency, rather than truncating at a fixed low order the way the papers' fixed-precision (20-significant-digit Maple) calculation could tolerate. Truncation order M is chosen so that successive terms no longer change the retained digits of the radial function at the working precision; no fixed M is hardcoded independent of dps.

The in-plane problem is analogous but vector-valued: the coupled radial functions F, G expand as

$$F(r^*, \zeta) = \sum_{m=0}^M a_m^{*(q)}(\zeta) r^{*\lambda(q)+m}, \quad G(r^*, \zeta) = \sum_{m=0}^M b_m^{*(q)}(\zeta) r^{*\lambda(q)+m} \quad (\text{Ref. [2] Eqs. (28)–(29)}),$$

with indicial equations

$$a_0^* \bar{c}_{11}^* \bar{r}_0 \lambda(\lambda - 1) = 0, \quad b_0^* \bar{r}_0 \lambda(\lambda - 1) = 0 \quad (\text{Ref. [2] Eq. (30a,b); } \bar{c}_{11}^* = c_{11}^*/c_{66}),$$

giving $\lambda = 0, 1$ (each a double root of the coupled system) and four coupled solutions selected by the uncoupled leading-coefficient scheme of Ref. [2] Eq. (32a–d): the two solutions built on $\lambda = 0$ differ in which of F, G carries the non-vanishing leading coefficient, and likewise for $\lambda = 1$, giving four independent, non-degenerate coupled radial functions overall.

In both the out-of-plane and in-plane cases, the transverse wavenumber (ξ or ζ) that indexes a physically admissible solution is a root of the radial-edge dispersion relation, and as the frequency Ω varies this root traces a curve that passes through real, imaginary, and complex regimes (Ref. [1] Fig. 2 plots several such branches). The branch-census construction tracks every branch continuously across frequency and re-fills any family a purely local root-finder would drop at a branch collision, so that the retained basis at every Ω contains one representative from each of the four families with no silent gaps, the practically difficult part of the replication, independent of the recursion detail above.

S.2 Free-edge surface-term algebra

The cantilever’s two corner terms (Eq. (43)). The tail of Ref. [1] Eq. (43), confirmed directly against the published page rather than inferred, reads

$$+(T - \bar{\nu}) \left[\frac{\partial^2}{\partial r^* \partial \theta} \left\{ \frac{w}{r^* + \bar{r}_0} \right\} \right]_{r^* = -\pi/2, \theta = -\Theta}^{r^* = \pi/2, \theta = -\Theta} - 2(T - \bar{\nu}) \left[\frac{\partial^2}{\partial r^* \partial \theta} \left\{ \frac{w}{r^* + \bar{r}_0} \right\} \right]_{r^* = -\pi/2, \theta = \Theta}^{r^* = \pi/2, \theta = \Theta} = 0, \quad (\text{B.1})$$

appended to an integral term (over $r^* \in [-\pi/2, \pi/2]$, evaluated at $\theta = -\Theta$) that enforces the clamped edge’s kinematic conditions in mixed/Lagrangian form, and a boundary term at $\theta = \Theta$ that gives the free edge’s own moment condition. Both displayed corner terms are jump contributions over the two radial edges $r^* = \pm\pi/2$ (i.e., $r = R_o$ minus $r = R_i$), but at a single, different circumferential edge apiece: the $+(T - \bar{\nu})$ term at the *clamped* edge $\theta = -\Theta$, and the $-2(T - \bar{\nu})$ term at the *free* edge $\theta = \Theta$. The two coefficients are not a sign-flipped pair of one functional: the clamped-edge term is generated by the paper’s substitution enforcing $w = \partial w / \partial \theta = 0$ along $\theta = -\Theta$ (a constraint on the primal solution, entering the variational statement through the same Lagrangian-substitution mechanism as the integral term above it), while the free-edge term is a genuine natural boundary contribution from the free edge’s own variation. Consequently the free-free extension’s second free edge ($\theta = -\Theta$, now released rather than clamped) must reuse the *free-edge* formula (the $-2(T - \bar{\nu})$ structure, correctly re-oriented) and cannot reuse Eq. (43)’s $+(T - \bar{\nu})$ coefficient, which is specific to a kinematically clamped edge and has no meaning once that edge’s kinematic constraint is removed.

The asymmetry is not an arbitrary bookkeeping choice: it has a classical-plate-theory origin. The free edge’s factor of 2 is the same doubling that produces the familiar concentrated corner reaction $2M_{r\theta}$ at a free corner in Kirchhoff plate theory, where the two adjoining free edges’ Kirchhoff shear conditions combine into a single term; the clamped edge has no analogous free-variation contribution to double, since its kinematic condition removes that degree of freedom from the variation entirely, leaving the single-multiplier Lagrangian term instead. The link from Ref. [1]’s already-published general functional (Eq. (6)) to this specific free-free result is derived explicitly, not asserted by analogy, in the algebra below; main-text §3.3 states the resulting free-free corner term and proves the same-parity cancellation of the mis-oriented assembly. What is genuinely

not reproduced here, and is a different and larger undertaking than the above, is an independent line-by-line integration-by-parts derivation of Eq. (6)/Eq. (1a) itself starting from the governing equation: that is Ref. [1]’s own foundational derivation, already published and peer-reviewed, and identical for the cantilever and free-free cases alike, so redoing it here would not add evidence specific to the free-free extension.

The intermediate algebra that produces (B.3) and (B.3’) from Ref. [1] Eq. (6) follows.

Grounding the corner-term asymmetry in the general functional (Eq. (6)). Ref. [1] Eq. (6), confirmed directly against the published page rather than inferred, is the true pre-specialization variational statement from which Eqs. (42)–(43) are later obtained by substituting the cantilever’s specific edge conditions:

$$\begin{aligned}
& \int_S \left[\left\{ \frac{\partial \tau_{rz}^{(0)}}{\partial r} + \frac{1}{r} \left(\frac{\partial \tau_{\theta z}^{(0)}}{\partial \theta} + \tau_{rz}^{(0)} \right) - 2\rho h \ddot{w} \right\} \delta w + \left\{ \frac{\partial \tau_{rr}^{(1)}}{\partial r} + \frac{1}{r} \frac{\partial \tau_{r\theta}^{(1)}}{\partial \theta} + \frac{\tau_{rr}^{(1)} - \tau_{\theta\theta}^{(1)}}{r} - \tau_{rz}^{(0)} \right\} \delta u_r^{(1)} \right. \\
& + \left. \left\{ \frac{\partial \tau_{r\theta}^{(1)}}{\partial r} + \frac{1}{r} \left(\frac{\partial \tau_{\theta\theta}^{(1)}}{\partial \theta} + 2\tau_{r\theta}^{(1)} \right) - \tau_{\theta z}^{(0)} \right\} \delta u_\theta^{(1)} \right] dS - \int_{-\Theta}^{\Theta} \left[r \left\{ \left(\tau_{rz}^{(0)} + \frac{1}{r} \frac{\partial \tau_{r\theta}^{(1)}}{\partial \theta} \right) \delta w - \tau_{rr}^{(1)} \delta \left(\frac{\partial w}{\partial r} \right) \right\} \right]_{r=R_i}^{r=R_o} d\theta \\
& - \int_{R_i}^{R_o} \left[w \delta \tau_{\theta z}^{(0)} - \frac{\partial w}{\partial r} \delta \tau_{r\theta}^{(1)} \right]_{\theta=-\Theta} d r - \int_{R_i}^{R_o} \left[\left(\tau_{\theta z}^{(0)} + \frac{\partial \tau_{\theta r}^{(1)}}{\partial r} \right) \delta w - \tau_{\theta\theta}^{(1)} \delta \left(\frac{1}{r} \frac{\partial w}{\partial \theta} \right) \right]_{\theta=\Theta} d r \\
& - \left[\tau_{r\theta}^{(1)} \delta w \right]_{r=R_i, \theta=-\Theta}^{r=R_o, \theta=-\Theta} + 2 \left[\tau_{r\theta}^{(1)} \delta w \right]_{r=R_i, \theta=\Theta}^{r=R_o, \theta=\Theta} = 0.
\end{aligned} \tag{B.2, \equiv \text{Ref. [1] Eq. (6)}}$$

Immediately after Eq. (6), Ref. [1] states that its last two surface terms were obtained “by performing the line integral around the contour C_N in Eq. (1a) along the three free edges” – i.e., Eq. (6) as printed is *already* built for a specific edge assignment (both radial edges plus $\theta = \Theta$ free; $\theta = -\Theta$ not among the “three free edges”), not a symmetric statement later specialized equally at both circumferential edges. This is visible directly in the two circumferential boundary terms’ structure: the $\theta = \Theta$ term is a stress (dual) quantity multiplying a variation of the primal deflection (δw , $\delta(\frac{1}{r} \partial w / \partial \theta)$) – the natural form whose stationarity under arbitrary δw gives the free edge’s own moment and shear conditions – while the $\theta = -\Theta$ term instead multiplies the primal deflection (w , $\partial w / \partial r$) against a variation of the dual/stress quantities ($\delta \tau_{\theta z}^{(0)}$, $\delta \tau_{r\theta}^{(1)}$), the mixed/Lagrangian form whose stationarity under arbitrary $\delta \tau$ instead constrains the primal kinematic quantities themselves. This is the level at which Eq. (B.1)’s coefficient asymmetry actually originates: not in Eq. (43)’s corner coefficients as such, but in Eq. (6) treating $\theta = -\Theta$ and $\theta = \Theta$ asymmetrically from the outset, before any constitutive substitution.

For the free-free extension, releasing $\theta = -\Theta$ from its clamped condition means its Eq. (6) boundary term must be replaced by the natural form. This substitution is carried out here explicitly from Eq. (1a)’s general indicial contour term $n_a \tau_{a\beta}^{(1)} s_\beta$, rather than asserted by pattern-matching, by transforming it into r, θ components at $\theta = \Theta$ and $\theta = -\Theta$ and tracking the outward-normal/tangent reversal between them through a counterclockwise traversal of the sector boundary starting at $(R_i, -\Theta)$: the $\theta = -\Theta$ edge is traversed with increasing r (tangent $\hat{s} = +\hat{r}$, outward normal $\hat{n} = -\hat{\theta}$), the $\theta = \Theta$ edge with decreasing r (tangent $\hat{s} = -\hat{r}$, outward normal $\hat{n} = +\hat{\theta}$). At $\theta = \Theta$ ($n = +\hat{\theta}$, $s = -\hat{r}$): $n_a t_{a3}^{(0)} = \tau_{\theta z}^{(0)}$, $u_{3,n}^{(0)} = \frac{1}{r} \partial w / \partial \theta$, $n_a t_{a\beta}^{(1)} s_\beta = -\tau_{\theta r}^{(1)}$, so $(n_a t_{a\beta}^{(1)} s_\beta)_{,s} = \partial \tau_{\theta r}^{(1)} / \partial r$ (the s -derivative along $-\hat{r}$ flips the sign of the r -derivative), and $n_a t_{a\beta}^{(1)} n_\beta = \tau_{\theta\theta}^{(1)}$; substituting into Eq. (1a)’s third term and converting $ds \rightarrow dr$ (tangent direction and integration limits both reverse, canceling) reproduces Eq. (6)’s printed $\theta = \Theta$ bracket exactly, leading minus sign included – a working self-consistency check before the method is trusted at the untested edge. At $\theta = -\Theta$ ($n = -\hat{\theta}$, $s = +\hat{r}$): $n_a t_{a3}^{(0)} = -\tau_{\theta z}^{(0)}$, $u_{3,n}^{(0)} = -\frac{1}{r} \partial w / \partial \theta$, $n_a t_{a\beta}^{(1)} s_\beta = -\tau_{\theta r}^{(1)}$ (the same value as at Θ ,

since for a symmetric tensor $n_a \tau_{a\beta} s_\beta$ is invariant under simultaneously flipping both $n \rightarrow -n$ and $s \rightarrow -s$ – a useful independent check), so $(n_a t_{a\beta}^{(1)} s_\beta)_{,s} = -\partial \tau_{\theta r}^{(1)} / \partial r$ (no sign flip this time, since $s = +\hat{r}$ already matches increasing r), and $n_a t_{a\beta}^{(1)} n_\beta = \tau_{\theta\theta}^{(1)}$ (two minus signs cancel). The integrand becomes the *negative* of the $\theta = \Theta$ bracket, and with $ds = +dr$ (no limit-flip this time) Eq. (1a)’s leading minus combines with this internal minus to give an overall plus: the *same natural-edge bracket* as the $\theta = \Theta$ term, with an *opposite overall sign*,

$$+ \int_{R_i}^{R_o} \left[\left(\tau_{\theta z}^{(0)} + \frac{\partial \tau_{\theta r}^{(1)}}{\partial r} \right) \delta w - \tau_{\theta\theta}^{(1)} \delta \left(\frac{1}{r} \frac{\partial w}{\partial \theta} \right) \right]_{\theta=-\Theta} dr, \quad (\text{B.3})$$

i.e. Eq. (6)’s $\theta = \pm\Theta$ terms become $[(\cdot)\delta w - (\cdot)\delta(\cdot)]_{-\Theta}^{\Theta}$ in the Eq. (7) bracket sense. This sign-alternation is confirmed two independent ways: by the (n, s) -orientation bookkeeping above, and by the assembled free-edge integral, which uses the same stress/displacement bracket on every edge with orientation $+1$ at $\theta = \Theta$ and -1 at $\theta = -\Theta$: exactly “same bracket, opposite sign.”

Fully substituting Eq. (B.3) into explicit deflection terms uses the same two constitutive relations Ref. [1] applies at $\theta = \Theta$ (Eq. (9c) and Eq. (15) enter only the clamped edge’s superseded *mixed*-term substitution, not this one), transcribed directly from the source (Ref. [1] pp. 763–764):

$$\begin{aligned} \tau_{\theta\theta}^{(1)} &= -\bar{D} \left\{ \bar{\nu} \frac{\partial^2 w}{\partial r^2} + \frac{R^*}{r} \left(\frac{\partial w}{\partial r} + \frac{1}{r} \frac{\partial^2 w}{\partial \theta^2} \right) \right\}, & (\text{Ref. [1] Eq. (14)}) \\ \tau_{\theta z}^{(0)} + \frac{\partial \tau_{\theta r}^{(1)}}{\partial r} &= -\bar{D} \left\{ \frac{R^*}{r^3} \frac{\partial^3 w}{\partial \theta^3} - \frac{(2T - R^*)}{r^2} \frac{\partial^2 w}{\partial r \partial \theta} + \frac{2T}{r} \left(\frac{\partial^3 w}{\partial r^2 \partial \theta} + \frac{1}{r^2} \frac{\partial w}{\partial \theta} \right) \right. \\ &\quad \left. - \bar{\nu} \left(\frac{1}{r} \frac{\partial^3 w}{\partial r^2 \partial \theta} + \frac{2}{r^3} \frac{\partial w}{\partial \theta} - \frac{2}{r^2} \frac{\partial^2 w}{\partial r \partial \theta} \right) \right\}, & (\text{Ref. [1] Eq. (20)}) \end{aligned}$$

together with the nondimensionalization $r = (2b/\pi)(r^* + \bar{r}_0)$ (Ref. [1] Eqs. (26)–(28)), so $\partial/\partial r = (\pi/2b) \partial/\partial r^*$ and $1/r = (\pi/2b)/(r^* + \bar{r}_0)$ ($T, R^*, \bar{\nu}$ are pure material-constant ratios, unaffected by this substitution; θ itself is not rescaled). Substituting Eqs. (14) and (20) into the $\theta = \Theta$ bracket and factoring reproduces Ref. [1]’s own published $\theta = \Theta$ block of Eq. (43) term-for-term (a self-consistency check with no reference to that printed result used in the substitution itself), and, since Eqs. (14) and (20) hold pointwise at any (r, θ) with no reference to Θ or $-\Theta$, the identical algebra applies verbatim at $\theta = -\Theta$ with only Eq. (B.3)’s outer sign flipped. The complete, explicit result, combining both edges in the Eq. (7) bracket sense, is

$$\begin{aligned} &\int_{-\pi/2}^{\pi/2} dr^* \left[\frac{1}{r^* + \bar{r}_0} \frac{\partial}{\partial \theta} \left\{ \frac{2Tw + R^* \partial_\theta^2 w}{(r^* + \bar{r}_0)^2} + 2T \partial_{r^*}^2 w - \frac{(2T - R^*)}{r^* + \bar{r}_0} \partial_{r^*} w \right. \right. \\ &\quad \left. \left. - \bar{\nu} \left\{ \frac{2w}{(r^* + \bar{r}_0)^2} + (r^* + \bar{r}_0)^2 \partial_{r^*} \left(\frac{\partial_{r^*} w}{(r^* + \bar{r}_0)^2} \right) \right\} \right\} \delta w \right. \\ &\quad \left. - \left\{ \frac{R^*}{r^* + \bar{r}_0} \left(\frac{\partial_\theta^2 w}{(r^* + \bar{r}_0)^2} + \frac{\partial_{r^*} w}{r^* + \bar{r}_0} \right) + \frac{\bar{\nu}}{r^* + \bar{r}_0} \partial_{r^*}^2 w \right\} \delta(\partial_\theta w) \right]_{\theta=-\Theta}^{\theta=\Theta} = 0, \quad (\text{B.3}') \end{aligned}$$

every term reached by explicit substitution of Eqs. (14), (20) and the nondimensionalization above at both edges, not reused at $\theta = -\Theta$ by sign-flip analogy from the published $\theta = \Theta$ block.

Closed-contour corner jump. For the free-free case the contour C_N is the full closed boundary, and the corner coefficient follows from Eq. (1a)’s fourth term, $\int_{C_N} ds (n_a \tau_{a\beta}^{(1)} s_\beta \delta w)_{,s}$, an exact differential along arc length whose segment-by-segment integration over a closed contour does not

vanish, because n_a, s_b are discontinuous at each corner even though the physical stress field $\tau_{r\theta}^{(1)}$ is continuous there. Writing $F \equiv n_a \tau_{a\beta}^{(1)} s_\beta$ (using each segment's own (n, s) convention) and summing the fundamental-theorem contributions corner by corner for the four segments traversed counterclockwise from $(R_i, -\Theta)$:

segment	tangent $\hat{s}$	normal $\hat{n}$	$F = n_a \tau_{a\beta}^{(1)} s_\beta$
$r = R_o$ arc ($\theta : -\Theta \rightarrow \Theta$)	$+\hat{\theta}$	$+\hat{r}$	$+\tau_{r\theta}^{(1)}$
$\theta = \Theta$ radial ($r : R_o \rightarrow R_i$)	$-\hat{r}$	$+\hat{\theta}$	$-\tau_{r\theta}^{(1)}$
$r = R_i$ arc ($\theta : \Theta \rightarrow -\Theta$)	$-\hat{\theta}$	$-\hat{r}$	$+\tau_{r\theta}^{(1)}$
$\theta = -\Theta$ radial ($r : R_i \rightarrow R_o$)	$+\hat{r}$	$-\hat{\theta}$	$-\tau_{r\theta}^{(1)}$

and taking the jump at each corner as (incoming segment's F) minus (outgoing segment's F),

$$\begin{aligned}
(R_o, -\Theta) : (-\tau_{r\theta}) - (\tau_{r\theta}) &= -2\tau_{r\theta}|_{(R_o, -\Theta)}, & (R_o, \Theta) : (\tau_{r\theta}) - (-\tau_{r\theta}) &= +2\tau_{r\theta}|_{(R_o, \Theta)}, \\
(R_i, \Theta) : (-\tau_{r\theta}) - (\tau_{r\theta}) &= -2\tau_{r\theta}|_{(R_i, \Theta)}, & (R_i, -\Theta) : (\tau_{r\theta}) - (-\tau_{r\theta}) &= +2\tau_{r\theta}|_{(R_i, -\Theta)}.
\end{aligned}$$

Summing the $\theta = \Theta$ pair reproduces Ref. [1]'s own printed cantilever corner term exactly, $(R_o, \Theta) + (R_i, \Theta) = 2[\tau_{r\theta}^{(1)} \delta w]_{R_i, \Theta}^{R_o, \Theta}$, a self-consistency check with no reference to the deployed code; and summing the $\theta = -\Theta$ pair, now smooth-turn corners of the closed contour rather than the cantilever's open ends, gives $(R_o, -\Theta) + (R_i, -\Theta) = -2[\tau_{r\theta}^{(1)} \delta w]_{R_i, -\Theta}^{R_o, -\Theta}$: magnitude 2 at every corner, alternating in sign between the two edge pairs. Substituting $\tau_{r\theta}^{(1)} = -\bar{D}(T - \bar{\nu})\partial^2(w/r)/\partial r \partial \theta$ (Ref. [1] Eq. (15)) and the same nondimensionalization used for Eq. (B.3') produces the free-free analogue of Eq. (43) stated in main-text §3.3.

Independent confirmation via Green's identity. The closed-contour jump just given fixes the corner term's magnitude (2, every corner) and its alternation pattern between the two edge pairs, but not, by itself, which edge carries the “+” label in an isolated equation, an overall sign-convention choice rather than physical content. A third, code- and FE-independent route settles it: a Green's-identity argument on the internal virtual-work bilinear form, evaluated on the classical constant pure-twist field ($M_{r\theta} = \text{const}$, $M_{rr} = M_{\theta\theta} = 0$, which satisfies the free-edge natural conditions identically), fixes the corner force to $R = 2M_{ns}$ with its sign read directly from the outgoing edge under $\partial\Omega$'s standard positive (counterclockwise) orientation; applied to the (n, s) table above, it reproduces the free-edge magnitude and the factor of 2. Magnitude and alternation therefore follow from the contour and Green arguments. Which isolated edge carries the “+” label is a convention of the overall stationarity equation — multiplying it by -1 swaps both labels and moves no frequency — and what is required instead, namely that the corner term and the surface term carry the same ± 1 orientation factor within one equation, is what the deployed assembly implements (main-text §3.3).

The free-free corner term is therefore constrained three ways (the closed-contour jump of this section, the Green's-identity argument just above, and comparison against the deployed assembly) rather than asserted by pattern-matching against Eq. (43) alone. The term-by-term factoring behind Eq. (B.3') is the algebra of this section. The same-parity cancellation of the mis-oriented assembly is proved in main-text §3.3.

S.3 Additional validation tables

The tables and supporting accounts below are referenced from main-text §4–§6 but collected here so the main validation narrative can stay on the tables that carry the paper’s primary claims. §S.3.6 is the raw-to-physical conversion algebra; §S.3.7 is the SUBDOM numerical procedure; Table S.8 is the flexural spurious-zero listing moved from the main text.

S.3.1 Basis-size convergence, per mode

The main text’s §4 summary states that 24 of 34 tabulated modes change by less than 0.1% across $n = 20, 28, 36$, with the remainder traced to two understood mechanisms. Full detail: nine labels (twelve (label, n) cells, three labels flagged at two basis sizes each) initially returned a value 0.6–12% away from their own seed at one basis size; re-polishing exactly those cells inside a bracket chosen to exclude the specific neighboring value the wide-window search had locked onto recovered, in every case, a value within 0.5% of the seed (cells marked [†] below use the recovered value). Two further cells, IP-11 at $n=20$ and IP-26 at $n=28$, sat just below the diagnostic’s original 1% flag threshold and were identified only by a later review pass; their tight-bracket re-polish recovered values within 0.03% of the seed. Separately, a follow-up diagnostic campaign established that the branch-selection step feeding this table frequently does not reach the labeled basis size: the routine orders candidate branches by $(|\operatorname{Im} \xi|, |\xi|)$ and fills greedily until n generalized coordinates are reached, which for this geometry’s branch inventory can leave an even-valued shortfall unfilled. Forcing the routine to reach the labeled n anyway was tested against all 34 modes at both $n=28$ and $n=36$, evaluated at each mode’s own production frequency: in every one of 32 such evaluations where this produced a larger basis than the routine’s own choice (29 at the modes’ initial seed frequencies, and a further 3 re-confirmed at their exact converged frequencies), $\sigma_{\min}$ degraded from a genuine, deep minimum to a shallow wide-window lock-on, confirming the shortfall is the routine correctly declining to admit a matrix-corrupting branch, not a defect. At $n=28$, all eight out-of-plane labels reach the full labeled basis; the in-plane labels mostly reach 26 of 28, with IP-05/IP-06 reaching only 22. At $n=36$, no label reaches the full labeled basis (out-of-plane: 28 or 32; in-plane: 30 or 34); the “used” columns below give the actually achieved size per row. IP-06’s own $n=36$ entry was recorded as 30 in one job and 34 in another, which an initial reading took for run-to-run irreproducibility. It is not: the achieved size is a deterministic but rapidly varying function of Ω in this neighborhood. Five repeat evaluations at the tabulated frequency return 30 every time, and a scan across $\Omega \in [1.196, 1.205]$ returns 30 except inside a band roughly 0.001 wide near $\Omega \approx 1.2032\text{--}1.2040$, where a tenth transverse-wavenumber branch enters the search window and, being a complex-quadrant pair worth four generalized coordinates, lifts the fill from 30 to 34.¹ The two values are therefore both correct, for marginally different frequencies; the range is retained in the table rather than collapsed. Consequently the $\Delta_{28 \rightarrow 36}$ columns compare two sub-maximal bases in every row, and the $\Delta_{20 \rightarrow 28}$ column for IP-05/IP-06 is a 20-to-22, not 20-to-28, comparison (flagged individually below). The basis size actually achieved by basis fill at each mode’s own converged Ω^* , tabulated as the “used” columns below, is confirmed by jobs 2324817/2324822.

A non-default split/anchor enhancement. Main-text §4 notes that a fully-default detector pass missed 4 of 26 tabulated extensional modes, later recovered as genuine roots of the

¹Jobs 2411851 and 2414031; the second covers $[1.199, 1.205]$, which the first did not reach. Transitions $30 \rightarrow 34$ and back occur between consecutive 2×10^{-4} scan points, so the two earlier jobs’ slightly different converged Ω placed them on opposite sides of one.

Table S.1: Basis-size convergence, out-of-plane modes (raw Ω ; labels OOP-01–OOP-08 correspond by row order to the main text’s Table 6, which does not print these labels). $\dagger$: recovered via a tight-bracket re-polish after an initial wide-window lock-on (see text above). “Used” columns: the basis size actually achieved by basis fill at the labeled target, confirmed at each mode’s own converged Ω^* .

Label	$n=20$	$n=28$	$n=36$	used@28	used@36	$\Delta_{20 \rightarrow 28}(\%)$	$\Delta_{28 \rightarrow 36}(\%)$
OOP-01	0.383601	0.383598	0.383646	28	28	−0.0007	+0.0125
OOP-02	0.616726 $\dagger$	0.616724	0.616751	28	28	−0.0003	+0.0044
OOP-03	0.961242 $\dagger$	0.961210	0.961210	28	32	−0.0033	+0.0000
OOP-04	1.369616	1.369586	1.370489	28	28	−0.0022	+0.0659
OOP-05	1.752160	1.752037	1.755174	28	28	−0.0070	+0.1791
OOP-06	2.320817	2.320723	2.320714	28	32	−0.0041	−0.0004
OOP-07	2.379839	2.379688	2.379688	28	28	−0.0063	+0.0000
OOP-08	2.602593	2.602553	2.602553	28	28	−0.0015	+0.0000

Table S.2: Basis-size convergence, in-plane modes (raw Ω ; labels IP-01–IP-26 correspond by row order to the main text’s Table 7, which does not print these labels). $\dagger$: recovered via a tight-bracket re-polish after an initial wide-window lock-on (see text above). “Used” columns: as Table S.1. $\ddagger$: IP-06’s used@36 is 30 at the tabulated frequency and 34 inside a narrow adjacent Ω band; both jobs are correct for their own converged Ω (see text). $\S$: $\Delta_{20 \rightarrow 28}$ computed on a 22-, not 28-coordinate basis: a $20 \rightarrow 22$ comparison, not $20 \rightarrow 28$.

Label	$n=20$	$n=28$	$n=36$	used@28	used@36	$\Delta_{20 \rightarrow 28}(\%)$	$\Delta_{28 \rightarrow 36}(\%)$
IP-01	0.401200 $\dagger$	0.401185	0.401185	26	30	−0.0037	+0.0000
IP-02	0.687385	0.687377	0.687377	26	30	−0.0011	+0.0000
IP-03	0.764608	0.764588	0.764580	26	30	−0.0026	−0.0010
IP-04	1.142356 $\dagger$	1.143136 $\dagger$	1.142313	26	30	+0.0682	−0.0720
IP-05	1.145249	1.144393	1.145192	22	30	−0.0747 $\S$	+0.0698
IP-06	1.197860	1.197592 $\dagger$	1.203379	22	30–34 $\ddagger$	−0.0224 $\S$	+0.4833
IP-07	1.416707	1.416366	1.416355	26	34	−0.0241	−0.0008
IP-08	1.440521	1.440118	1.440110	26	34	−0.0280	−0.0006
IP-09	1.523138	1.522316	1.522258	26	34	−0.0540	−0.0038
IP-10	1.598919	1.598834	1.598826	26	34	−0.0053	−0.0005
IP-11	1.613504 $\dagger$	1.613132	1.613057	26	34	−0.0230	−0.0046
IP-12	1.690176	1.690020	1.689972	26	34	−0.0092	−0.0029
IP-13	1.779739 $\dagger$	1.779680	1.779676	26	34	−0.0033	−0.0002
IP-14	1.832690	1.831850	1.831788	26	34	−0.0458	−0.0034
IP-15	1.974997	1.972955	1.972920	26	34	−0.1034	−0.0018
IP-16	1.987420	1.987180	1.987158	26	34	−0.0121	−0.0011
IP-17	1.997673	1.996470	1.996243	26	34	−0.0602	−0.0114
IP-18	2.169765	2.166997	2.166775	26	34	−0.1276	−0.0102
IP-19	2.294238 $\dagger$	2.293218 $\dagger$	2.293042	26	34	−0.0445	−0.0077
IP-20	2.302741	2.301518	2.304771	26	34	−0.0531	+0.1413
IP-21	2.389179	2.388598	2.388507	26	34	−0.0243	−0.0038
IP-22	2.456152	2.450795	2.450496	26	34	−0.2181	−0.0122
IP-23	2.524864	2.521487	2.521288	26	34	−0.1337	−0.0079
IP-24	2.570691	2.566361	2.565772	26	34	−0.1684	−0.0230
IP-25	2.679963	2.676029	2.674815	26	34	−0.1468	−0.0454
IP-26	2.777863	2.772007 $\dagger$	2.771646 $\dagger$	26	34	−0.2108	−0.0130

same $n = 28$ determinant. A candidate enhancement to the blind pipeline itself, an opt-in local split/anchor/adaptive-bisection strategy that re-examines a coarse candidate’s own local zoom for more than one distinct minimum before committing to a single bracket, was tested in the same whole-range, no-seeding configuration and recovered all four of those modes unaided, including the closest pair (~ 0.0043 apart in Ω). The same run left a different two-mode pair unrecovered under

the 0.1% match tolerance (~ 0.0011 apart, tighter than the fine-zoom step itself); a targeted follow-up dense scan subsequently confirmed both are genuine, distinct roots. The enhancement therefore recovered 24 of 26 tabulated modes. It is not the package default: a required second-geometry blind pass (in-plane orthotropic) recovered only 1 of 8 known targets, and escalating the search basis left the same mixed pattern, so a basis-size difference is not the cause. Job-level detail: §S.4.

S.3.2 Instrument-by-instrument adjudication of the near-degenerate ODD pair

Main-text §6.3 states the outcome of the eight-instrument suite applied to the two candidates (raw $\Omega = 2.828013$ and 2.838515) against the single FE target at $\Omega_{\text{lit}} = 111.46$. The individual instruments and their outcomes are:

1. *Basis-size bump* ($n_{\text{dofs}} 20 \rightarrow 28$): both candidates persist, essentially unchanged.
2. *Sensitivity map*: a fixed-offset scan around each candidate’s golden-section result found no evidence that either was locking onto a nearby artifact rather than a true root.
3. *Null-vector gap analysis*: no informative separation.
4. *Parity readout*: at the calibrated basis sizes both read cleanly ODD, self-parity correlation 1.0000, matching the FE target.
5. *ODD-block topology scan*: no structural difference between the two.
6. *Modal assurance criterion*: the decisive instrument, and the only one that compares mode *shape* rather than frequency or matrix structure. MAC against the FE eigenvector is 0.9913 for 2.828013 and 0.9999 for 2.838515, a gap of 0.0086 against the 0.2 forced-call threshold. Both candidates sit in the “good correlation” range of experimental modal practice [3] (MAC > 0.9; its uncorrelated band is < 0.05), so the absolute-level convention does not separate them either, and the observed gap is roughly $20\times$ too small to force a call at the gap threshold used here.
7. *Pure-slope (moment-slope) residual*, applied independently after the first six were exhausted: the two candidates’ residual ratios differ by $6.6\times$ (2.549×10^{-4} against 1.682×10^{-3}) and their phase-aligned null-vector cosine is only 0.228, indicating two radially distinct modes rather than one being a numerical echo of the other.
8. *Enlarged-basis check* ($n_{\text{ret}} = 34/38$, beyond the calibrated $n_{\text{dofs}} = 20/28$): candidate 2.828013’s self-parity reads ODD at $n_{\text{ret}} = 34$ and EVEN at 38. Follow-up probes traced this to a redistribution of null-vector energy onto a different dominant root (from a pure-imaginary root carrying 31% of the energy to a complex root carrying 76%) and ruled out a basis-selection artifact directly: removing the single marginal root that crosses the basis-fill boundary changes retained energy by under 0.1% and leaves the parity unchanged.

MAC/FE nodal correspondence. Instrument 6 above and the two-sided extensional bar of main-text §6.6 both compare the exact solution’s reconstructed mode shape against an FE eigenvector by MAC, and both build the comparison the same way. The exact reconstruction is not evaluated directly at the FE node coordinates; it is first evaluated on a dense regular grid in the solver’s own natural coordinates (r mapped affinely onto $[-\pi/2, \pi/2]$, θ over $[-\Theta, \Theta]$, bisector at $\theta = 0$). Each FE node’s angular coordinate (measured from the FE model’s own origin at one circumferential edge, not from the bisector) is shifted onto the solver’s bisector-centered convention

before use. The grid’s radial and transverse displacements are then bilinearly interpolated onto every FE node’s coordinates; a node whose coordinates fall outside the grid’s covered domain returns a non-finite value and is excluded from that particular MAC evaluation rather than extrapolated. The interpolated radial/transverse pair is rotated into the FE model’s own Cartesian (U_X, U_Y) using each node’s own global angle, so the MAC correlation of main-text §6.6 is taken entirely in the FE model’s native frame.

A separate correspondence step is needed only to classify an FE eigenvector’s own symmetry class (SYMM/ANTI about the bisector), which the two-sided bar’s class-safe matching requires so a candidate is never compared against an opposite-parity FE target. This pairs each FE node with its own mirror-image node across the bisector, within the same mesh, and is independent of the exact-solution correspondence above. At the half-annulus ($2\Theta = \pi$, the §6.5 geometry) this pairing is exact: a node’s reflected polar coordinates coincide with another printed node’s coordinates to within rounding. At every other sector angle, including the primary quarter-annulus $2\Theta = \pi/2$ and the rest of the main-text §6.4 sweep, the FE decks are meshed in Cartesian coordinates, so a node’s exact polar mirror image generally does not land on another node’s printed coordinates. The production fallback, used at those angles, is a nearest-neighbor spatial search in Cartesian coordinates for the closest node to the exact mirror image, accepted only within half a typical element edge length; a node with no partner inside that radius is dropped from the symmetry calculation rather than mis-paired. The primary-geometry MAC calibration of main-text §6.6 therefore used this fallback, not exact pairing. At the half-annulus, where exact pairing succeeds, forcing the Cartesian fallback instead left all 57 FE symmetry labels and all eight calibration MAC calls unchanged at threshold 0.744.

Owning-block scoring and held-out candidates. Two properties of the extensional reconstruction have to be respected before MAC means anything. The parity blocks $q=0$ and $q=1$ of the assembled K are the sector’s two exact symmetry classes, so a MAC taken across them is analytically zero: a reading of 0.0000 records a block-selection outcome, not an absence of correlation. And the smallest singular vector of a *non*-singular block varies smoothly with Ω , so a block evaluated near a neighboring candidate’s root reproduces that neighbor’s mode shape: one candidate at the primary geometry reads 0.9658 this way without owning a root in that block at all. Each candidate must therefore be scored on the parity block that owns a root at its own Ω .

Selecting that block requires a rule, and no single criterion suffices. Choosing the more singular block misclassifies one candidate at the second geometry; choosing the smaller edge residual misclassifies a different one at the first. Requiring that a block’s $\log_{10} \sigma_{\min}$ sit at a local minimum in Ω before it is eligible, and breaking ties on the residual, classifies 24 of 24 against 23 of 24 for either criterion alone. At (2.0, 0.5) all three selection criteria score 8 of 8, so that geometry confirms the bar transfers but supplies no evidence that the rule is better than the simpler alternatives; the rule’s margin rests on two candidates.

Two candidates were held out of the calibration set entirely, because the pointwise residual and the FE evidence disagree about them. At $\Omega = 1.667187$ the production residual verdict is ARTIFACT-like, the independent FE frequency match is 0.011%, and MAC on the root-owning block reads 0.9981; at $\Omega = 2.117877$ the corresponding figures are 0.006% and 0.9980. Both sit far above any threshold in the window, so MAC agrees with the FE evidence where the residual does not. This hold-out pair is not the eight residual-unresolved candidates of main-text §6.5: those eight include $\Omega = 1.667187$ as one of two MAC-REAL calls (the other is $\Omega = 1.894176$, MAC 0.9953); $\Omega = 2.117877$ is a calibration hold-out only. Main-text §6.6 records the production call (midpoint 0.744 of the $\nu = 0.30$ window); §6.5 records the outcome on the eight.

The in-plane analogue of Table S.8, listing owning-block residuals for these eight calibration candidates, is Table S.7.²

Calibration-window robustness. Main-text §6.6 quotes the seed-based REAL floor and ARTIFACT ceiling (0.9079, 0.5801) because they require no extra refinement step. Evaluating the one calibration candidate whose catalogued frequency lies furthest from its own converged root at that root instead raises the combined REAL floor from 0.9079 to 0.9869. The combined window (ARTIFACT ceiling 0.5801 to REAL floor) therefore widens from 0.3278 to 0.4068; the per-geometry gap at (2.0, 1.0) on the seed-based table is already that same 0.4068 and is unchanged to the printed digits. The reported window is otherwise insensitive to how the FE comparison target is chosen: MAC taken against each candidate’s own nearest-frequency FE target gives the same window as MAC taken as the best correlation over every sampled target, so the calibration does not depend on the nearest-frequency pairing being correct. Unrestricted nearest-frequency matching is nevertheless not class-safe on its own: at $(r_0/2b, 2\Theta/\pi) = (2.0, 1.0)$ one calibration artifact’s nearest-frequency FE target belongs to the opposite symmetry class, and MAC against that target is identically zero, the wrong-class signature described above, not an absence of correlation. This is why main-text §6.6 scores every candidate as the maximum over FE modes of the same symmetry class as the root-owning block, not simply the nearest frequency.

S.3.3 Cut-off-adjacent screening and the extensional basis-packing shortfall

Main-text §6.4 sets aside a cut-off-adjacent subset of the 24-geometry (48 geometry/motion) sweep and reports only its final tallies. That subset was screened with the same §5 pointwise discriminator used for the primary tables, its first application outside the single geometry at which it was calibrated. All 119 flexural candidates were resolved to their target basis and adjudicated: 57 REAL-like, 62 ARTIFACT-like, 0 AMBIGUOUS (a 47.9% REAL-like fraction; these 119 are a subset of the 695 flexural candidates screened in main-text §6.4, whose complement reads 272/535, or 50.8%).³ The REAL-like and ARTIFACT-like populations do not overlap ($r_M \leq 0.041$ against $r_M \geq 2.05$ for the directly reconstructed majority, a ≥ 50 -fold separation, narrower than, but qualitatively identical to, the primary geometry’s own ~ 2000 -fold gap).

The extensional side has no calibrated *residual* verdict bar at any geometry, so its figures are raw residual ratios. All 96 extensional candidates were resolved: 88 bimodal, 8 middle-band by the same $r_{Tyy} < 0.3 / > 4$ split used at the primary geometry (r_{Tyy} and r_{Tyr} here are the main text’s extensional residual ratios $r_{\theta\theta}$ and $r_{\theta r}$, in the solver’s own edge-frame naming). Seventy-one reached their target basis (direct search plus an escalation ladder); the remaining 25 were stuck at exactly $n_{\text{dofs}} - 2$ even at a guarded x_{max} cap of 90 and were evaluated at that achieved, smaller basis, following the same used@28/used@36 relabeling precedent as §S.3.1 and the IP-06 footnote, with one exception: a candidate that reached the labeled fill with a shallow σ_{min} (−2.681) and is reported at that fill.⁴

The 25-candidate shortfall comes from a packing artifact of greedy branch selection; the underlying inventory is not shrinking. Each admitted root contributes either 2 degrees of freedom (real or pure-imaginary) or 4 (a complex-quadrant pair); the routine fills by ascending ($|\text{Im } \zeta|, |\zeta|$) and stops when the next candidate would overshoot the target. At each shortfall point the running fill reaches exactly $n_{\text{dofs}} - 2$ with only 4-dof pairs remaining. The same cap arises at a control geometry where

²Job 2434754. Owning-block residuals match the calibration residuals to 10^{-6} relative.

³Direct pass: jobs 2416866–2416869. Escalation ladder for the 12 that did not reconstruct on a cold search: job 2417234.

⁴Job 2419006.

30–34 degrees of freedom are available, so a genuine geometry-dependent drop in complex-branch inventory with increasing $r_0/2b$ coexists but is not the cause. No adjudicated (flexural) conclusion depends on this. The full-sweep match recapture behind Table S.3 is job 2416746 (superseding an intermediate figure from an earlier recapture of the same sweep, 1231 of 1474, 83.5%, mean 0.578%, worst 3.000%, which had itself already superseded the original interactively-run, never-committed statistic, 1232 of 1466, 84.0%; the per-candidate set is not identical across any of these checkpoints, and the aggregate figure is what each recapture re-derives). That job also settles how much the one unrecoverable methodological choice matters: rerun at extensional-contamination tolerances of 0.5%, 1.0% and 2.0%, it returns 1261, 1256 and 1253 matches (85.5%, 85.2%, 84.9%), with mean-of-per-geometry-means 0.573%/0.574%/0.574% and worst case 2.990% in all three — a fourfold change in the unrecorded parameter moves the headline by 0.6 percentage points. The reported figures are the 1.0% column. The flexural residual screen behind its reconstruction/verdict tallies is jobs 2428355–2428358.

Table S.3: Free-free detector accuracy vs. an independent FE benchmark, 4 radius ratios $\times$ 6 sector angles (both motion types), excluding each geometry’s cut-off-adjacent candidates; main-text §6.4. n : matches within 3% of 1475 attempted. The sweep produces 2291 candidates in all (695 flexural, 1596 extensional); 1475 remain after removing each geometry’s cut-off-adjacent candidates (119 flexural, 96 extensional) and those lying beyond the FE extraction range (12 flexural, 589 extensional). Every accepted flexural candidate was residual-screened: 654 of 695 reconstructed, 329 REAL-like / 321 ARTIFACT-like / 4 AMBIGUOUS. Cut-off-adjacent subset and extensional packing shortfall: §S.3.3.

$r_0/2b$	R_i/R_o	n	mean err (%)	max err (%)
1.5	0.500	274	0.572	2.976
1.66667	0.538	294	0.559	2.944
2.0	0.600	336	0.576	2.965
2.5	0.667	352	0.590	2.990
all	—	1256/1475	0.574	2.990

The $\nu = 0.35$ flexural candidate $\Omega = 1.432743$ in Table S.4 below, which job 2347329 could not reconstruct at $n=16$, was recovered by the cut-off-adjacent escalation ladder (job 2421415) as ARTIFACT-like and FE-matched within 3%; the table’s IP FE-matched rates use the 80-mode extract (jobs 2421055/2421056).

Table S.4: Second-geometry adjudication summary ($r_0/2b = 1.5$, $2\Theta/\pi = 1.0$), both motion types, both materials; main-text §6.5. FE-matched: within 3% of an independent FE frequency. The $\nu = 0.35$ flexural candidate $\Omega = 1.432743$ was recovered by the cut-off-adjacent escalation ladder as ARTIFACT-like and FE-matched within 3%; it is included. IP FE-matched rates use the 80-mode extract.

ν	motion	N	REAL-like	ARTIFACT-like	AMBIG.	FE-matched
0.35	flexural (OOP)	28	14	14	0	21/28 (75.0%)
0.35	extensional (IP)*	61	31	19	11	57/61 (93.4%)
0.30	flexural (OOP)	30	15	15	0	20/30 (66.7%)
0.30	extensional (IP)*	64	28	18	18	61/64 (95.3%)

*raw residual-ratio classification, not a calibrated residual verdict (main-text §6.3; mode-shape bar: §6.6).

Table S.5: Second radius-ratio adjudication summary ($r_0/2b = 2.0$, $R_i/R_o = 0.600$, $2\Theta/\pi = 1.0$, $\nu = 0.30$); main-text §6.5. FE-matched: within 3% of an independent FE frequency.

ν	motion	N	REAL-like	ARTIFACT-like	AMBIG.	FE-matched
0.30	flexural (OOP)	33	18	15	0	25/33 (75.8%)
0.30	extensional (IP)*	75	35	18	22	68/75 (90.7%)

*raw residual-ratio classification, not a calibrated residual verdict (main-text §6.3; mode-shape bar: §6.6).

Table S.6: Crossed-geometry adjudication summary ($r_0/2b = 2.0$, $R_i/R_o = 0.600$, $2\Theta/\pi = 0.5$, $\nu = 0.30$); main-text §6.5. FE-matched: within 3% of an independent FE frequency. All 10 flexural REAL-like candidates are among the 11 FE-matched; the eleventh is the ARTIFACT-like FE-mode-16 companion discussed in main-text §6.5.

ν	motion	N	REAL-like	ARTIFACT-like	AMBIG.	FE-matched
0.30	flexural (OOP)	18	10	8	0	11/18 (61.1%)
0.30	extensional (IP)*	47	16	16	15	43/47 (91.5%)

*raw residual-ratio classification, not a calibrated residual verdict (main-text §6.3; mode-shape bar: §6.6).

Table S.7: Owning-block traction residuals at $n=20$ for the primary-geometry MAC calibration set of main-text §6.6 (the listing analogue of Table S.8). Labels are that calibration assignment, not a row-by-row application of the §5 $\sim 0.3/\sim 4$ split. The REAL-like half are main-text Table 7 rows 1, 4, 10 and 23; the ARTIFACT-like half are catalogued spurious zeros of the same determinant. Residuals are on the root-owning parity block (local-minimum residual rule); the last numeric column is $\max(r_{Tyy}, r_{Tyr})$.

Label	Ω	block	r_{Tyy}	r_{Tyr}	$\max(r_{Tyy}, r_{Tyr})$	Verdict
real_a	0.401154	$q=1$	0.03052	0.01808	0.03052	REAL-like
real_b	1.142287	$q=0$	0.04982	0.03534	0.04982	REAL-like
real_c	1.598985	$q=1$	0.04855	0.02166	0.04855	REAL-like
real_d	2.521489	$q=1$	0.6858	0.3205	0.6858	REAL-like
art_a	0.542105	$q=0$	6.907	4.801	6.907	ARTIFACT-like
art_b	1.021144	$q=1$	6.058	8.287	8.287	ARTIFACT-like
art_c	1.454616	$q=0$	7.035	3.288	7.035	ARTIFACT-like
art_d	1.603259	$q=0$	5.199	0.917	5.199	ARTIFACT-like

Table S.8: FF-P1 flexural spurious zeros ($n_{\text{dofs}} = 20$), the listing referred to from main-text §6.3. Depth is $\log_{10} \sigma_{\min}$ at the same basis. Physical modes at this basis have r_M spanning 0.0025–0.037 and depths -2.81 to -4.78 (main-text Table 6). The last row, raw $\Omega = 2.817613$, is main-text Fig. 2(e).

Raw Ω	Parity	r_V	r_M	depth	Verdict
0.320670	EVEN	53.4	74.9	-3.69	ARTIFACT-like
0.844488	ODD	75.2	76.0	-4.25	ARTIFACT-like
1.001419	EVEN	454	290	-5.29	ARTIFACT-like
1.602090	EVEN	95.6	75.5	-4.71	ARTIFACT-like
2.121685	ODD	620	348	-4.13	ARTIFACT-like
2.817613	EVEN	180.6	75.52	-4.82	ARTIFACT-like

S.3.4 Extensional close-pair resolutions

Two extensional benchmark pairs are close enough in frequency that a single root is asked to cover both FE or literature targets, for opposite reasons. Main-text §6.3 and §6.9 state only the outcome; both are resolved here in full.

The 1114.9/1115.3 Hz pair (main-text §6.3, Table 7). Isolated golden-section refinement at the two predicted pair members, in brackets that cannot reach each other, finds two shift-stable deep minima: $\Omega^* = 2.772422$ (1115.18 Hz, depth -4.43) and $\Omega^* = 2.773399$ (1115.58 Hz, depth -4.82), each 0.015% from one FE member. The production detector still returns a single singularity (2.772007 at $n=28$; 2.772924 at $n=36$, between the two isolates). MAC against the two isolates assigns both REAL-like to the same FE mode (FE#29, MAC 0.9898 and 0.9901); the production root $\Omega = 2.771955$ is that same shape (MAC 0.9896). One shape, two frequencies; the main-text table is unchanged.

Qin’s 4.4293/4.4754 pair (main-text §6.9, Table 10). This is not Table 7’s FE-matched pair (rows 4–5, $\Omega = 1.142287$ and 1.145158 , 0.25% apart at the nominal wavenumber window). Those two roots are the comparison against FE in main-text §6.3; they are not an assignment of Qin’s published values, and no claim is made about which spacing is correct. The $n_{\text{dofs}} = 12$ search lists a single nearby root (1.147615) against both published Qin targets. Repeating that targeted search at requested $n=20$ and $n=28$ — an enlarged x_{max} search that realizes 18 degrees of freedom whichever of the two is requested — still returns one singularity ($\Omega = 1.14274$, $\Omega_{\text{Qin}} = 4.4529$), 0.53% and 0.50% from the two Qin targets. A second isolated dip sits 0.42% away at $\Omega = 1.14758$ ($\Omega_{\text{Qin}} = 4.4717$, depth -5.18 at $n=20$ and -6.07 at $n=28$, shift 0.0001%); the Qin-targeted search still returns only the 1.14274 root, so Qin’s pair remains unseparated. Isolation brackets of $\pm 0.15\%$ cannot reach the neighboring root; this is not a table split, the published row stays the $n=12$ shared listing. MAC on the two isolates assigns both REAL-like to distinct FE modes of opposite parity: $\Omega = 1.14274$ ANTI, MAC 0.9996 at FE#7 (Table 7’s 459.5 Hz partner; the isolate itself converts to 459.7 Hz); $\Omega = 1.14758$ SYMM, MAC 0.9871 at FE#8 (Table 7’s 460.6 Hz partner; the isolate itself converts to 461.6 Hz). Here the outcome runs the other way: MAC finds two distinct shapes, but the Qin-targeted search still returns only the one root.

The two pairs illustrate the same limit from opposite sides: one production root can legitimately stand for two nearby targets when their mode shapes coincide (the 1114.9/1115.3 Hz pair), or, when they do not (Qin’s pair), a targeted search still cannot assign a distinct production root to each published Qin value even though MAC shows the two isolates are physically distinct shapes.

S.3.5 Shi et al. out-of-plane refinement detail

Main-text §6.10 states only the final refined roots (Table 11); two of Shi et al.’s six targets needed more than the basic window-scan-plus-golden-section pipeline, detailed here. The acceptance criteria applied throughout both cases are $\log_{10} \sigma_{\text{min}} \leq -3.5$, error below 2%, and both bracket-shift replicas within 0.1% of the converged root.

53.649: a genuine near-degenerate pair. The initial $\pm 2.5\%$ -bracket golden-section refinement failed its bracket-shift acceptance test (the refined position moved 0.59% when the bracket was nudged), a symptom main-text §5’s instruments treat as lock-on instead of a located root. An unconstrained dense scan of the neighborhood isolated two genuine, stable roots 0.59% apart: $\Omega^* = 1.957111$ ($\Omega_{\text{lit}} = 53.6553$, err 0.012%, depth -5.848) at the published frequency, and $\Omega^* = 1.945645$

($\Omega_{\text{lit}} = 53.341$, err 0.574%, depth -3.861), the wide bracket’s lock-on target. Each root is stable to 0.0004% or better inside an isolated $\pm 0.4\%$ bracket that cannot reach the other, confirming the original 0.59% walk was the separation of two roots, not an unlocated singularity. The published value is reported in Table 11 against the deeper, closer member; the second root stands on its own as a genuine singularity, left unforced to the published value.

38.428: a basis-depth near-miss, not a null result. At the production screening basis $n=20$, a dip sits exactly on the predicted frequency, 1.9 decades below the local background (deepest point $\log_{10} \sigma_{\text{min}} = -3.297$ at raw $\Omega = 1.40170$), but falls short of the -3.5 acceptance depth. A 33-point scan of the tight $\pm 5\%$ window confirmed this is a genuine depth shortfall, not a scan-resolution miss (the unrelated feature 12.5% away that won the original wide window’s arg-max lies outside this tighter window). Repeating at $n=28/36$ clears the acceptance bar (-3.823 , -4.312); golden-section refinement then gives a shift-stable root at $\Omega_{\text{lit}} = 38.4295$ (err 0.004%, depth -4.706 at $n=28$; $n=36$ agrees to 38.4289, depth -5.166), reported in the table as a refined root.

S.3.6 Raw-to-physical conversion constants

The isotropic free-free benchmark underlying main-text Tables 6 and 7 (both the SHELL281 flexural and the PLANE183 extensional FE models) is a single physical plate: inner radius $R_i = 4$ m, outer radius $R_o = 8$ m (so $R_i/R_o = 0.5$ and $r_0/2b = 1.5$), thickness $H = 0.08$ m, sector angle $2\Theta = \pi/2$ (half-angle $\Theta = \pi/4$, the quarter-annulus of main-text §6.2–§6.3), steel material $E = 210$ GPa, $\nu = 0.30$, $\rho = 7800$ kg/m³. The governing slenderness is thickness over the smallest in-plane dimension, the radial width $2b = 4$ m, giving $H/2b = 0.02$ ($H/R_o = 0.01$) — well inside the range where the classical thin-plate model applies, though see main-text §6.3 on the residual theory difference against a shear-deformable finite-element benchmark. With $b = (R_o - R_i)/2 = 2$ m, $c_{66} = G = E/2(1 + \nu)$ and $\bar{c}_{11} = E/(1 - \nu^2)$, the two nondimensionalizing constants entering main-text §3.1’s $\bar{\Omega} = \omega/\bar{\omega}$ (Ref. [1] Eqs. (26)–(28); Ref. [2] Eqs. (23)–(25)) are

$$\bar{\omega} = \frac{\pi}{2b} \sqrt{\frac{c_{66}}{\rho}} = 2527.351 \text{ rad/s}, \quad \bar{k} = \frac{2b}{\pi} \sqrt{\frac{3c_{66}}{h^2 \bar{c}_{11}}} = 32.617, \quad h = \frac{H}{2}.$$

The raw solver output converts to physical Hertz directly for the in-plane (extensional) tables, $f = \bar{\Omega}\bar{\omega}/2\pi$, verified against the FEM entries in main-text Table 7, confirming the in-plane raw column already *is* $\bar{\Omega} = \omega/\bar{\omega}$ with no further conversion needed. The out-of-plane raw column is *not* $\bar{\Omega}$: $\bar{\Omega}$ enters the flexural dimensionless equation only through the product $\bar{k}\bar{\Omega}$, which is what Ref. [1]’s Table 2 and main-text Table 6 both report. Recovering physical frequency from the flexural raw column therefore carries one extra division by $\bar{k}$,

$$f_{\text{OOP}} = \frac{(\bar{k}\bar{\Omega})\bar{\omega}}{2\pi \bar{k}},$$

and this is what populates Table 6’s $f(\text{Hz})$ column, giving both flexural and extensional tables a shared physical-frequency metric even though their raw columns are different quantities. Omitting the $\bar{k}$ here would inflate every flexural frequency by a factor of 32.617 for this plate and would place the first flexural mode above the first extensional one, which is not physical for a plate of these proportions. The isotropic out-of-plane table (main-text Table 6) is, in its principal frequency column, reported in the classical literature convention $\Omega_{\text{lit}} = \omega R_o^2 \sqrt{\rho H/D}$ ($D = EH^3/12(1 - \nu^2)$), the convention Shi, Liang, Wang and Teng and the original Seok–Tiersten tables both use), which

is linear in the row column $\bar{k}\bar{\Omega}$ through the same factor $\bar{k}$:

$$\Omega_{\text{lit}} = (\bar{k}\bar{\Omega}) \frac{\bar{\omega}}{\bar{k}} R_o^2 \sqrt{\frac{\rho H}{D}} = \omega R_o^2 \sqrt{\frac{\rho H}{D}}.$$

Table 5 uses the same Ω_{lit} symbol under the orthotropic constants of main-text §3.4, not this isotropic $\bar{k}$ conversion. The conversion is cross-checked numerically against all 8 rows of main-text Table 6 (e.g. raw 0.383695 $\rightarrow \Omega_{\text{lit}} = 15.148$, raw 0.961312 $\rightarrow 37.951$, raw 2.602500 $\rightarrow 102.743$, each matching the tabulated solver value to the last printed digit). The extra factor of $\bar{k}$ for the out-of-plane case (absent from the in-plane conversion) reflects that $\bar{\omega}$ alone nondimensionalizes the extensional (second-order, wave-speed-scaled) problem, while the flexural (fourth-order, thickness-scaled) problem’s frequency parameter additionally carries the thickness-dependent wavenumber scale $\bar{k}$; both constants derive from the same base quantities b, c_{66}, ρ (and, for $\bar{k}$, $h, \bar{c}_{11}$), so no new physical input is required beyond the geometry and material stated above.

S.3.7 SUBDOM: numerical procedure and gate detail

This section gives the computational detail behind main-text §6.7 at a level a reimplementaion could follow; the main text states only the construction and the results.

Factorization and whitening. For a fixed Ω , the in-plane free-free block assembly is $\mathbf{K} = \mathbf{D}^\top \mathbf{T}$ (bold denotes the matrices of this construction, distinguishing $\mathbf{D}, \mathbf{T}, \mathbf{M}, \mathbf{R}$ from the scalars $D, T, M_{r\theta}$ and R^* used elsewhere), verified entrywise against the deployed `_build_K_real` assembly on the same basis (gate G1, bar 10^{-30} , measured 1.21×10^{-41} to 8.80×10^{-41} across all geometries reported). $\mathbf{D}$ is QR-decomposed, $\mathbf{D} = \mathbf{Q}\mathbf{R}$; the pencil is whitened by forming $\mathbf{M} = \mathbf{T}\mathbf{R}^{-1}$; the singular value decomposition of $\mathbf{M}$ gives generalized singular values $\sigma_1 \leq \sigma_2 \leq \dots \leq \sigma_n$, of which $\sigma_1 = \eta(\Omega)$ and the associated right singular vector, mapped back through $\mathbf{R}^{-1}$, is the traction-minimizing coefficient vector.

Degeneracy pooling and interior projection. With $\gamma = 4$ (adopted from the labelled-set window $\gamma = 1.5\text{--}8$ measured before any blind run; $\gamma = 3.0$ was the original pre-registered value), $k = \#\{i : \sigma_i \leq G\sigma_1\}$ selects the near-degenerate cluster of lowest-traction directions. Each of the k associated coefficient vectors, together with the null vector c_{null} of the accepted zero, is mapped to an interior displacement field via `field_of()/adv.build_grid_from_cfull` rather than the edge trace (an edge-trace version is defeated by the mode-2 bleed-through artifact, main-text §6.7). The k interior fields are Gram–Schmidt orthonormalized under the same identity-weighted inner product used for the main-text §6.6 MAC, on the interior $r \times \theta$ grid those helpers produce; SUBDOM is the cumulative squared projection of the normalized c_{null} interior field onto that orthonormal span.

Gates, restated with bars. G1: entrywise $\mathbf{K} = \mathbf{D}^\top \mathbf{T}$, bar 10^{-30} . G2: reproduces the published FF-P1 extensional residual ratios, bar 10^{-4} ; applicable only at the primary geometry and production basis ($n_{\text{dofs}} = 20$, $x_{\text{max}} = 20$), reported not-applicable elsewhere by design. G3: the algebraic identity $\eta(\Omega) \leq \eta(c_{\text{null}})$, which must hold exactly since c_{null} is one feasible point of the minimization defining η . G4: cross-checks the edge-trace decomposition SUB1 (not SUBDOM itself) against an explicit MAC computation, bar 10^{-9} ; SUB1 and SUBDOM are expected to disagree in general, since one is evaluated on the edge trace and the other on the interior field. G5: confirms the candidate’s Ω is a genuine local minimum of σ_{min} under this basis, rejecting candidates that moved under the current search (reported `C.MOVED`, not a failure of the criterion). G6: negative-control

siting (known non-root points must fail G5 and must also read SUBDOM below threshold); not applicable in blind-mode runs, where no candidate is pre-labelled.

Calibration set. The threshold, $\text{SUBDOM} > 0.70$, was set at $n_{\text{dofs}} = 20$ as the midpoint of the window spanned by 52 labelled points across four (geometry, ν) combinations: the primary geometry, the two further $\nu = 0.30$ geometries of main-text §6.5, and the $\nu = 0.35$ geometry of main-text §6.4. The 52 points include the two historical near-degenerate “killer” modes (one of which, at an enlarged basis $n_{\text{dofs}} = 28/x_{\text{max}} = 60$, is the sole exception recorded against this criterion to date, characterized and not hidden) and 13 non-root negative controls (points where $K(\Omega)$ has no accepted zero at all). All 52 classify correctly at that cut. It was then frozen: 0.70 was fixed before any of the cluster runs in main-text §6.7 were submitted — the four-geometry curated confirmation and the 122-candidate blind set alike — and was not retuned afterward. “Pre-registered” below refers to that freeze, not to a value chosen before the calibration itself. $\gamma = 4$ was adopted after a second-geometry blind set showed $\gamma = 2$ –6 inert and after the sole $\gamma = 3.0$ knife-edge (FE mode 23, $\gamma = 3.01$) had been identified; it is a value inside the labelled-set window. A re-score of the stored blind checkpoints at $\gamma = 4$ is 45/49 and 25/28 with zero duplicates accepted (at $\gamma = 3.0$, 44/49 and 25/28).

Block selection at near-degenerate pairs. Where two solver structures sit within $\lesssim 10^{-3}$ in Ω (as at $r_0/2b = 2.0$, $2\Theta/\pi = 1.0$ ’s FE mode 35, main-text §6.7), the production pipeline’s `RULE_LOCALMIN_RESID` rule selects which parity block to reconstruct by which is locally more singular — inherited from main-text §6.6’s own pipeline, not independently re-derived for SUBDOM. This is the same style of automatic sigma-based selection whose unmodified reuse, in the standalone diagnostic probe built to cross-check the FE mode 35 pairing, silently reconstructed the same block for both candidates on its first attempt (an exact $\text{MAC} = 0.0000/0.0000$ result for both, later traced to the bug and not reported as a finding). The fix there — an explicit `force.block` parameter with an in-run fidelity gate proving `force.block=None` reproduces the original automatic selection bit-for-bit — is specific to that standalone probe; the production pipeline’s own `RULE_LOCALMIN_RESID` rule is unmodified. Forcing both blocks at the remaining disputed modes (main-text §6.7) shows that this inherited rule, not the interior projection, is what misses FE modes 32 and 26 at $n_{\text{dofs}} = 20$, $x_{\text{max}} = 20$: the auto-picked block reads $\text{MAC} = 0.0000$, while the other block at the same Ω matches the frequency-nearest FE mode ($\text{MAC} = 0.8528$ and 0.9302) and, at FE mode 32, $\text{SUBDOM} = 0.7136$ on that matching block, above the pre-registered cut (FE mode 26’s matching block reads $\text{SUBDOM} = 0.8728$). FE mode 26 recovers at an enlarged basis, below (main text §6.7); FE mode 32 does not. FE mode 50 had no owner above 0.744 in FE modes 4–60 under this first pass (best $\text{MAC} = 0.4963$) and neither block certified ($\text{SUBDOM} = 0.0894/0.4813$); an Ω scan and golden-section refinement subsequently located a second, previously uncatalogued structure at $\Omega = 2.144291$ ($\text{SUBDOM} = 0.9974$, $\text{MAC} = 0.9993$) that resolves this pairing (main text §6.7). FE mode 30’s auto-picked block is a pairing artifact; the other block reads $\text{SUBDOM} = 0.7942$ but MAC against FE#30 and against the eleven existing r200/a50 eigenvectors is below 0.3 — max-over-blocks would accept it, and is not adopted.

A fourth geometry’s MAC cross-check. The $\nu = 0.35$ geometry ($r_0/2b = 1.5$, $2\Theta/\pi = 1.0$) in the main-text SUBDOM calibration table had SUBDOM values only when that table was first populated: no ANSYS eigenvector extract existed for it at the time. An 80-mode PLANE183 extract (job 2433080, modes 4–60 exported as mesh96 eigenvectors) has since made the same SUBDOM-vs-MAC bake-off used at the other three geometries (main-text §6.6’s calibration; jobs 2447412–

2447414) available here too. Run in the diagnostic container against all 12 of this geometry’s labelled points ($n_{\text{dofs}} = 20$, $x_{\text{max}} = 14$, matching the SUBDOM table), with `MAC_XMAX = 14` set explicitly before loading the reconstruction helper (the module-load-time default-binding pitfall documented above for `XMAX0`): all 6 REAL-labelled points read $\text{MAC} > 0.999$ on the SUBDOM-selected block; all 6 ARTIFACT-labelled points reject on that block, three below $\text{MAC} = 0.3$ and three via $\text{SUBDOM} \leq 0.70$ while their MAC sits in the 0.3–0.744 ambiguous band. One ARTIFACT candidate ($\Omega = 0.424067$) shows the same non-selected-block bleed-through already documented for `ffp1`’s `art.d`: its other block independently reads $\text{MAC} = 0.7699$, $\text{SUBDOM} = 0.8343$, both above their respective bars — confirmation, not a new finding, that max-over-blocks is not a safe alternative rule at this geometry either.

S.3.8 Cantilever replication: basis-size convergence and in-plane finite-element check

The main text’s cantilever tables (§6.1) are computed at the production basis sizes ($n = 20$ for $2\Theta/\pi \leq 0.5$; otherwise $n = 16$ out-of-plane and $n = 14$ in-plane).

Out-of-plane rows. Table S.9 re-runs every out-of-plane row at larger bases. For each row, a fresh transverse-wavenumber search is done at the printed frequency. At each basis size the production fill rule then selects the basis, which is tracked in Ω , and $\log_{10} \sigma_{\text{min}}$ of the boundary matrix is minimised by golden section over $[0.99, 1.01]$ times the printed value. At the production size this fixed-basis re-run reproduces every printed value to within 0.03%. “Last step” is the change between the two largest bases; a value above about 0.05% means the row is still moving at the largest basis tried.

The clamped edge converges more slowly with basis size than the free edges of the main study. Across the 20 rows the printed values move by 0.01–0.90% (median 0.26%). Every row moves upward except $2\Theta/\pi = 0.25$, mode 2. Each mode jumps once, when the transverse branch that shapes it enters the basis, and then plateaus. The converged values agree with [1] to within 2.14%. The largest residual is the narrowest sector’s fundamental, which is still changing by 0.07% per four added coordinates at $n = 44$; every other row is within 1.51%. The published values are themselves finite-truncation results, so a converged row’s residual measures both truncations together, not an error of either.

In-plane rows. The same procedure is not used in-plane. There the natural frequencies are double roots of the unconstrained determinant (main text §3.3). At a finite basis a double root separates into two nearby simple roots $A < B$ of $\det \mathbf{K}$, or into a complex-conjugate pair that leaves only a shallow real-axis minimum of σ_{min} . A golden-section search in a fixed window then settles on one member of the pair, or on the shallow minimum, and which one depends on the window as much as on the basis. The in-plane rows are therefore checked against finite elements (Table S.10).

The FE model is ANSYS PLANE183 (eight-node plane-stress quadrilaterals) on the same plate: $R_i = 3$, $R_o = 7$ ($r_0/(2b) = 1.25$), $E = 210$ GPa, $\nu = 0.35$, $\rho = 7800$ kg m^{−3}, wall $\theta = 0$ clamped and the other three edges free. $\bar{\Omega} = \omega/\bar{\omega}$ with $\bar{\omega} = (\pi/2b)\sqrt{c_{66}/\rho}$. Each geometry is solved on three mapped meshes (48^2 , 96^2 and 192^2 elements, radial by circumferential, with the circumferential count doubled at $2\Theta/\pi = 1.0$), and the tabulated value is the Richardson limit of the three. The observed order is about 1.5, set by the stress singularity at the clamped–free corners. The change from the 96^2 to the 192^2 mesh is at most 0.003%. An independent model of the same element order (nine-node Lagrange quadrilaterals with the exact polar geometry map, meshes 16^2 to 64^2 ,

Table S.9: Out-of-plane cantilever replication (main-text Table 3): basis-size convergence of every row. n_p : production basis size. Printed: the main-text value (at n_p). Converged: value at the largest basis $n_{\max}$ tried. Last step: change between the two largest bases (%). Err: signed deviation from [1] (%).

$r_0/(2b)$	$2\Theta/\pi$	Mode	n_p	Printed	Converged	$n_{\max}$	Last step	Err printed	Err conv.
1.25	0.25	1	20	0.345324	0.348425	44	+0.070	+1.23	+2.14
1.25	0.25	2	20	1.018530	1.016853	44	-0.001	+0.50	+0.34
1.25	0.25	3	20	1.722434	1.724811	44	+0.008	+0.22	+0.35
1.25	0.50	1	20	0.099419	0.099942	44	+0.036	+0.36	+0.89
1.25	0.50	2	20	0.343742	0.345096	44	+0.024	+0.90	+1.30
1.25	0.50	3	20	0.605904	0.605980	32	+0.004	+0.28	+0.29
1.25	0.75	1	16	0.050717	0.050906	28	+0.018	-0.22	+0.15
1.25	0.75	2	16	0.151458	0.152134	28	-0.001	+1.06	+1.51
1.25	0.75	3	16	0.376503	0.376825	28	+0.008	+0.06	+0.14
1.25	1.00	1	16	0.033153	0.033279	28	+0.023	-0.20	+0.18
1.25	1.00	2	16	0.080358	0.080623	28	-0.004	+0.74	+1.07
1.25	1.00	3	16	0.243521	0.243723	28	+0.002	+0.16	+0.25
1.25	1.25	1	16	0.024652	0.024746	28	+0.021	+0.05	+0.43
1.25	1.25	2	16	0.049473	0.049595	28	+0.001	+0.27	+0.52
1.25	1.25	3	16	0.151555	0.151757	28	+0.000	+0.26	+0.39
1.25	1.50	1	16	0.019790	0.019854	28	+0.013	+0.30	+0.63
1.25	1.50	2	16	0.034523	0.034602	28	+0.008	-0.08	+0.15
1.25	1.50	3	16	0.097470	0.097629	28	+0.002	+0.22	+0.38
5/3	1.00	1	16	0.018023	0.018073	24	+0.004	-0.15	+0.13
5/2	1.00	1	16	0.007762	0.007776	28	+0.008	-0.10	+0.07

Richardson limit) reproduces these limits to within 0.001% on all twelve modes compared (the first four at each sector angle).

Table S.10: In-plane cantilever replication (main-text Table 4) against finite elements. Printed: production value at n_p . Paper: [2]. FE: PLANE183 Richardson limit. Δ : signed deviation from FE (%).

$2\Theta/\pi$	Mode	n_p	Printed	Paper	FE	Δ printed	Δ paper
0.25	1	20	0.341243	0.345890	0.348738	-2.15	-0.82
0.25	2	20	0.808929	0.819210	0.823991	-1.83	-0.58
0.25	3	20	0.940032	0.940120	0.940599	-0.06	-0.05
0.50	1	20	0.120677	0.120880	0.121449	-0.64	-0.47
0.50	2	20	0.347033	0.351910	0.353053	-1.71	-0.32
0.50	3	20	0.529239	0.529300	0.529881	-0.12	-0.11
1.00	1	14	0.039466	0.039480	0.039555	-0.23	-0.19
1.00	2	14	0.106529	0.105560	0.105825	+0.67	-0.25
1.00	3	14	0.263049	0.261290	0.261755	+0.49	-0.18

The published values of [2] lie 0.05–0.82% below FE on every row. The production values lie within 2.15% of FE. Table S.11 locates both members of the split pair directly for the two narrow-sector rows furthest from FE ($2\Theta/\pi = 0.25$, modes 1 and 2). The transverse-wavenumber inventory is first checked for completeness by an argument-principle count of the zeros of the dispersion determinant in $[0, 30] \times [0, 45]$ of the complex ζ plane, at $\Omega = 0.345$, 0.81 and 0.94 (13, 14 and 14 zeros, all found). The basis is then tracked in Ω , and every sign change of $\det \mathbf{K}$ is bisected. The FE frequency lies between A and B at every basis size, the separation shrinks as the basis grows, and the midpoint approaches FE. The printed values are the lower members A at $n = 20$. The Lagrange-constrained determinant of main text §3.3 has one root C near the pair. It is not uniformly closer to FE: for mode 1 at $n = 20$ it lies below FE, for mode 2 above. Mode 3 at $2\Theta/\pi = 0.25$ is a complex pair for $n = 20$ –28, so its printed value is the shallow minimum; that

minimum is 0.06% from FE. By $n = 36$ the pair is real, with its lower member in $[0.9395, 0.9400]$ and C in $[0.9405, 0.9410]$, on either side of FE. The $2\Theta/\pi = 0.5$, mode 2 production root (0.347033) is likewise the lower root: at $n = 20$, $C = 0.354458$ lies above FE, and $\det \mathbf{K}$ changes sign only once in $[0.343, 0.3565]$ at a 0.0015 step.

Table S.11: In-plane cantilever, $2\Theta/\pi = 0.25$: the two unconstrained roots $A < B$ and the constrained root C against basis size. Width: $(B - A)$ over the midpoint (%). Mid: signed deviation of $(A + B)/2$ from FE (%). FE: 0.348738 (mode 1), 0.823991 (mode 2).

Mode	n	A	C	B	Width	Mid	A vs FE
1	20	0.341240	0.347160	0.351363	2.92	-0.70	-2.15
1	32	0.346927	0.348182	0.349034	0.61	-0.22	-0.52
2	20	0.808931	0.832999	0.835380	3.22	-0.22	-1.83
2	24	0.814646	0.828283	0.829999	1.87	-0.20	-1.13
2	28	0.817167	0.826797	0.829142	1.45	-0.10	-0.83
2	32	0.819106	0.825618	0.827223	0.99	-0.10	-0.59
2	40	0.820851	—	0.825988	0.62	-0.07	-0.38

S.4 Cluster job-number provenance

The main text states every result in prose, with no job number in its own running text; this section is the audit trail for the handful of findings the main text also flags with a footnote marker but does not itself enumerate by job. Each entry below is keyed to the main-text location that cites this section.

- **§4, split/anchor/adaptive enhancement.** Job 2409241 recovered 24 of 26 tabulated modes, including the closest pair discussed in the main text; jobs 2410035/2410044 independently confirmed the two remaining misses as genuine, distinct roots. The required second-geometry blind pass (in-plane orthotropic), job 2410036, recovered only 1 of 8 known targets; a follow-up that escalated the search basis, job 2428927, left the same mixed pattern, ruling out a basis-size explanation.
- **§5, extensional residual portability.** Job 2421896 confirmed the flexural screen unchanged at $dps = 25/40/60$ on the primary and two further geometries. Job 2421904 showed the extensional residual ratios still overlap at $r_0/2b = 2.0$, the basis for the one-sided-only claim.
- **§5, FF-P1 own-root separation.** Jobs 2409212/2409213 evaluated the complete FF-P1 REAL-like/ARTIFACT-like set at each candidate’s own converged root at $n = 20$ and $n = 28$.
- **§6.1, in-plane cantilever finite elements.** Jobs 2530949 ($2\Theta/\pi = 0.5, 1.0$) and 2530984 ($2\Theta/\pi = 0.25$) ran the PLANE183 decks (`ansys/NewAnsys/ansys_p1_cant_ip_r125_a025_2026-09-24.inp` and its `a050`, `a100` companions). The nine-node model is `validation/probe_p1_cant_ip_q9fe_2026-09-24.py`; the split-pair root scans are in `validation/p1_cant_ip_2026-09-24/`.
- **§6.3, the 1114.9/1115.3 Hz pair.** Job 2428351 isolated the two golden-section minima in brackets that cannot reach each other; job 2434752 ran MAC against the two isolates.
- **§6.3, candidate 2.828013’s basis-sensitive self-parity.** Jobs 2408715, 2409032, 2409058.
- **§6.4, geometry-sweep recapture.** Job 2416746 is the full, current-SOLVER.VERSION recapture of all four sweep geometries behind Table S.3; jobs 2428355–2428358 are the flexural residual-screen pass over its 24 out-of-plane keys.

- **§6.5, FE mode 9 / FE#10 neighbor.** Job 2428350 re-derived the fidelity of the FE#10 neighbor (reproducing job 2410396’s depth); job 2410396 is the source per-candidate adjudication data for FE#23; job 2421895 is the unconstrained scan that found no second minimum for FE#23.
- **§6.5, crossed-geometry table.** Jobs 2421898/2421899 (solver capture), 2421900/2421901 (ANSYS extracts), 2422897 (merge), 2422898 (adjudication).
- **§6.6, MAC-bar calibration robustness.** Jobs 2431872, 2431873 and 2432008 produced the three-geometry calibration table; job 2432028 independently reproduced job 2432008 bit-for-bit against a transcribed gate table; job 2431917 re-evaluated the one off-root candidate at its own root instead of its catalogued seed.
- **§6.6, C-class 80-mode extract.** Jobs 2445267–2445274 rescored all eight $2\Theta/\pi = 1.25$ and 1.50 keys against an 80-elastic-mode PLANE183 extract (modes 4–83). Every key printed C80-C_SURVIVES: 0 tight MAC-REAL at threshold 0.744 (reconstructed 82–115/82–115 per key). The tabulated 24-key sweep remains the 57-mode Rank 14 extract; this pass only tests whether a deeper list creates a REAL side.
- **§6.7, SUBDOM cluster confirmation.** Job 2445746 scored the curated candidate lists at the three non-primary geometries (all A_CONFIRMS, zero B_OVERLAP); job 2444987 is the primary geometry’s own curated run (15/15, A_CONFIRMS) and the source of the two PROBE points that prospectively predicted the pairing-defect resolution below. Jobs 2445957/2445958 are the full blind-adjudication-set test, split across independent SLURM array tasks (8-way at $r_0/2b = 2.0$, $2\Theta/\pi = 1.0$, 75 candidates; 5-way at $r_0/2b = 2.0$, $2\Theta/\pi = 0.5$, 47 candidates; 122 total, the count cited in main-text §6.7) and combined by a dedicated merge script that recomputes each row’s verdict from its own stored gate record. The ANSYS extraction of FE mode 35’s eigenvector at 682.741 Hz (mesh64/mesh96, cross-converged MAC 0.9991) is unnumbered in this log (a cluster ANSYS job, not a solver probe). Job 2446007 is the first MAC cross-check attempt, later found to have a block-selection bug (both candidates silently reconstructed the same parity block); job 2446128 is the corrected re-run, with an in-run fidelity gate proving the fix reproduces the original unforced behavior bit-for-bit, giving the cited MAC = 0.9944/0.0000 result. Job 2446328 forced both blocks at the three Rank-8 leftovers (FE modes 32/50/53) against existing fullblock eigenvectors: FE 53 is a pairing artifact (both MAC below 0.3); FE 32’s other block matches at 0.8528; FE 50 is ambiguous (best nearby MAC 0.4963, job 2446900). A later Ω scan resolved FE 53 as a third pairing defect with a replacement root: job 2448570 found a SUBDOM-real local $\sigma_{\min}$ minimum on the matching parity block at $\Omega = 2.265637$; job 2448862 scored MAC = 0.9971 against FE 53 there; job 2448978 found the standard unconstrained golden-section primitive unsafe on this window (a non-matching-block structure dominates nearby); job 2449358 therefore ran a block-constrained golden section on the matching block’s own $\sigma_{\min}$, converging to $\Omega^* = 2.265896$; job 2449466 re-scored MAC = 0.9978 against FE 53 at that polished point and MAC = 0.0000 against the non-matching block. Job 2446899 scored SUBDOM on both blocks at FE 32’s Ω : the matching block reads 0.7136. Job 2446521 is the r200/a50 FE mode 16 MAC (both blocks below 0.3, pairing artifact). Job 2446902 is FE modes 26/30 at that geometry (FE 26 other-block MAC = 0.9302; FE 30 both below 0.3). Job 2446898 recomputed four labelled FF-P1 points at dps = 60; SUBDOM agrees with the dps = 40 table to the printed digits. Job 2446965 is a duplicate of 2446902. Jobs 2447415–2447418 recomputed the full labelled set at all four geometries at dps = 60 (unique checkpoints); SUBDOM matches the dps = 40 table to the printed digits. Job 2447410 found no FE owner

for leftover F6/FE mode 50 in modes 4–60 (best MAC 0.4963); job 2447861 scored both blocks SUBDOM = 0.0894/0.4813. Job 2448321 scanned Ω across this window and found a second structure near 2.144515; job 2448496 scored MAC = 0.9974 against FE#50 there; a golden-section refinement (matching `find_modes_sigmin`’s own Phase 2 primitive) polished it to $\Omega = 2.144291$ (SUBDOM = 0.9974, MAC = 0.9993 on recheck), resolving the pairing. A stored-data-only search (jobs 2446899, 2446902, 2447411, 2447412, 2445746; no new solves) found no block-pick rule, other than `RULE_LOCALMIN_RESID` itself or max-over-blocks, that reaches FE mode 32 or 26 without also certifying a known artifact (ffp1 candidate `art.d`) REAL; both remained open misses at $n_{\text{dofs}} = 20$. Jobs 2448222/2448223 then tested basis enlargement: FE mode 32 stays a clean miss at $n_{\text{dofs}} = 24$, $x_{\text{max}} = 40$ and 28/60 (its relocated well is an unrelated mode, MAC ≈ 0.06); FE mode 26 shows a basis-fragile signal at $n_{\text{dofs}} = 24$ (RELOCATED_REAL, $\Omega = 2.076435$) that does not reproduce at 28/60 within a ± 0.04 window. Job 2448320 widened the $n_{\text{dofs}} = 28$, $x_{\text{max}} = 60$ window to ± 0.10 and found a deeper well at $\Omega = 2.112435$ plus a second, shallower minimum nearby, leaving the shape (one valley or two) unresolved. Job 2448501 ran a finer 0.002-step connecting scan across both and resolved it: a single continuous σ_{min} valley with its true minimum at $\Omega = 2.110435$ (SUBDOM = 0.9985, MAC = 0.8919, a single clean qualifying local minimum). A golden-section refinement (`golden_refine_ip_r200a50_m26_v1.py`, same primitive and bracket-construction convention as the FE-50 refinement above) polished this to $\Omega = 2.110860724725\dots$, gated against an already-validated same-basis catalogue root (r200/a50 s_1 , drift 1.1×10^{-3} against a 5×10^{-3} bar) and against its own scan-grid estimate (shift 4.3×10^{-4} against a 2×10^{-3} bracket half-width). The MAC/SUBDOM recheck at this polished value first read MAC = 0.4578, a sharp, suspicious drop from the scan-grid’s 0.8919 over a 4×10^{-4} shift in Ω ; tracing it found that the recheck script’s load of `probe_ip_pairing_defect_r200a100_v2.py` had not set the `MAC_N_DOFS/MAC_XMAX` environment variables that module reads at import time to bind its own default basis, so the MAC reconstruction had silently fallen back to $n_{\text{dofs}} = 20$, $x_{\text{max}} = 20$ instead of the enlarged basis in use — the same class of import-time default-binding pitfall documented earlier in this section for `XMAX0`. Setting those variables before the module load, and confirming the fix reproduces job 2448501’s stored MAC = 0.8919 bit-for-bit at the identical scan-grid Ω before trusting the polished- Ω number, gives the corrected recheck: SUBDOM = 0.9991, MAC = 0.8906, a single clean qualifying local minimum, both comfortably above their respective bars – the cited FE mode 26 recovery (main text §6.7). Job 2447411 scored FE 26/30 SUBDOM on both blocks (q1 0.8728 at FE 26; q0 0.7942 at FE 30). Jobs 2447412–2447414 are the labelled SUBDOM-vs-MAC bake-off (REAL/HELD all same-block; $t_6/t_8/s_8$ ARTIFACT-ambiguous). Jobs 2447419/2447420/2447863 are the KILL2 $n = 28/x_{\text{max}} = 60$ scan: catalogue Ω remains a local minimum at SUBDOM = 0.1857; the neighbouring genuine root 1.987180 relocates to 1.981966 (SUBDOM = 0.9886). A stored-checkpoint re-score at $\gamma = 4$ is 45/49 and 25/28 with zero duplicates (jobs 2445957/2445958 rows). Jobs 2447871–2447874 are the labelled set at $n_{\text{dofs}} = 24$, $x_{\text{max}} = 40$: local-minimum points still separate at all four geometries (the in-container r200/a100 window is reproduced); r200/a50’s printed `B_OVERLAP` is a non-root negative control sitting on a well that exists at this basis, not a labelled-root miss. Jobs 2447875–2447878 are the labelled set at $n_{\text{dofs}} = 28$, $x_{\text{max}} = 60$: KILL2 remains a local minimum at SUBDOM = 0.1857; the other three geometries confirm on the points that are still roots. Jobs 2447879–2447882 repeat $n = 28$ at each geometry’s own production x_{max} (KILL2 is REAL at $x_{\text{max}} = 20$; the miss is the $x_{\text{max}} = 60$ setting). Jobs 2447883/2447884 are the full blind adjudication set at $n = 28/x_{\text{max}} = 60$, merged from the same array pattern as 2445957/2445958: 28/31 and 14/15 FE modes that still have a local-minimum claimant, zero duplicates accepted (many catalogue Ω are `C_MOVED`

and excluded). Job 2447885 extracted r200/a50 fullblock eigenvectors for modes 4–60 (57 files, mesh96); job 2447886 found FE mode 30’s non-picked block matches FE mode 28 at $\text{MAC} = 0.7709$, not FE mode 30. Job 2433080 is the 80-mode PLANE183 extract (modes 4–60, mesh96) supplying the fourth geometry’s own SUBDOM-vs-MAC bake-off, above; that bake-off itself ran in the diagnostic container, not as a separate SLURM job.

- **§6.8, Wang orthotropic table.** Job 2408719 (targets 5–8), resuming targets 1–4 from job 2408466; an in-run fidelity gate on the isotropic reduction was confirmed before either search ran.
- **§6.9, Qin table.** Job 2409260 is the original $n = 12$ search. Jobs 2421062/2421063 are the Qin-targeted $n = 20/28$ re-search with an enlarged $x_{\max}$, each realizing 18 degrees of freedom. Job 2421905 isolated the second nearby dip at $\Omega_{\text{Qin}} = 4.4717$. Job 2434753 ran MAC on the two isolated dips against their respective FE targets.
- **§6.10, Shi et al. out-of-plane table.** Job 2414020 produced the four directly refined roots and set the acceptance criteria (depth ≤ -3.5 , error below 2%, bracket-shift replicas within 0.1%) reused for the remaining two. Job 2421061 resolved the 53.649 entry’s 0.59% bracket-shift walk into two stable, genuine roots. Job 2411852 confirmed the 38.428 entry’s $n = 20$ near-miss was not a scan-resolution artifact. Job 2421581 refined 38.428 at $n = 28/36$ to a shift-stable root.

References

- [1] J. Seok, H.F. Tiersten, “Free vibrations of annular sector cantilever plates. Part 1: out-of-plane motion,” J. Sound Vib. 271 (2004) 757–772.
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- [3] R.J. Allemang, “The modal assurance criterion – twenty years of use and abuse,” Sound Vib. 37(8) (2003) 14–21.